(245) $\lim_{n\to\infty} \frac{n+1}{n} = \lim_{n\to\infty} \frac{\frac{n}{n}+\frac{1}{n}}{\frac{n}{n}} =$

$\lim_{n\to\infty} \frac{1+\frac{1}{n}}{1} = \lim_{n\to\infty} 1+\frac{1}{n} =$

$\lim_{n\to\infty} 1$

(246) $\lim_{n\to\infty} \frac{(n+1)^2}{2n^2} = \lim_{n\to\infty} \frac{\frac{n^2+2n\cdot 1+1^2}{n^2}}{\frac{2n^2}{n^2}} =$

$\lim_{n\to\infty} \frac{2n+1^2}{2} = \lim_{n\to\infty} \frac{2n+1}{2} =$

$\lim_{n\to\infty} \frac{1}{2}$

(247) $\lim_{n\to\infty} \frac{(n+1)^3-(n-1)^3}{(n+1)^2+(n-1)^2} =$

$\lim_{n\to\infty} \frac{(n^3+3n^2+3n+1)-(n^3-3n^2+3n+1)}{(n^2+2n+1)+(n^2-2n+1)} =$

$\lim_{n\to\infty} \frac{6n^2+2}{2n^2+2} = \lim_{n\to\infty} \frac{\frac{6n^2}{n^2}+\frac{2}{n^2}}{\frac{2n^2}{n^2}+\frac{2}{n^2}} =$

$$\lim_{n\to\infty} \frac{6+\frac{2}{n^2}}{2+\frac{2}{n^2}} = \lim_{n\to\infty} \frac{6}{2} = \lim_{n\to\infty} 3$$

(248) $$\lim_{n\to\infty} \frac{n^3-100n^2+1}{100n^2+15n} = \frac{\frac{n^3}{n^3}-\frac{100n^2}{n^3}+\frac{1}{n^3}}{\frac{100n^2}{n^3}+\frac{15n}{n^3}}$$

$$\lim_{n\to\infty} \frac{1-\frac{100}{n}+\frac{1}{n^3}}{\frac{100}{n}+\frac{15}{n^2}} = \lim_{n\to\infty} +\infty$$

(249) $$\lim_{n\to\infty} \frac{1000n^3+3n^2}{0{,}001n^4-100n^3+1} =$$

$$\lim_{n\to\infty} \frac{\frac{1000n^3}{n^4}+\frac{3n^2}{n^4}}{\frac{0{,}001n^4}{n^4}-\frac{100n^3}{n^4}+\frac{1}{n^4}} =$$

$$\lim_{n\to\infty} \frac{\frac{1000}{n}+\frac{3}{n^2}}{0{,}001-\frac{100}{n}+\frac{1}{n^4}} =$$

$$\lim_{n\to\infty} \frac{0+0}{0{,}001-0+0} = \lim_{n\to\infty} 0$$

(250) $\lim_{n\to\infty} \frac{(n+1)^4 - (n-1)^4}{(n+1)^4 + (n-1)^4} =$

$\lim_{n\to\infty} \frac{(n^4 + 4n^3 + 6n^2 + 4n + 1) - (n^4 - 4n^3 + 6n^2 - 4n + 1)}{(n^4 + 4n^3 + 6n^2 + 4n + 1) + (n^4 - 4n^3 + 6n^2 - 4n + 1)}$

$\lim_{n\to\infty} \frac{8n^3 + 8n}{2n^4 + 12n^2 + 2} = \lim_{n\to\infty} \frac{\frac{8n^3}{n^4} + \frac{8n}{n^4}}{\frac{2n^4}{n^4} + \frac{12n^2}{n^4} + \frac{2}{n^4}} =$

$\lim_{n\to\infty} \frac{\frac{8}{n} + \frac{8}{n^3}}{2 + \frac{12}{n^2} + \frac{2}{n^2}} = \lim_{n\to\infty} \frac{0+0}{2+0+0} =$

$\lim_{n\to\infty} 0$

(251) $\lim_{n\to\infty} \frac{(2n+1)^4 - (n-1)^4}{(2n+1)^4 + (n-1)^4} = \lim_{n\to\infty}$

$\frac{(2n^4 + 8n^3 + 12n^2 + 8n^3 + 1) - (n^4 - 4n^3 + 6n^2 - 4n^3 + 1)}{(2n^4 + 8n^3 + 12n^2 + 8n^3 + 1) + (n^4 - 4n^3 + 6n^2 - 4n^3 + 1)}$

$\lim_{n\to\infty} \frac{n^4 + 12n^3 + 6 + 12n^3}{3n^4 + n + 18n^2 + 4 + 2} = \lim_{n\to\infty} \frac{n^4 + 24n^3 + 6}{5n^4 + 10 + 18n^2}$

$$\lim_{n\to\infty} \frac{\infty + \infty + 6}{\infty + 10 + \infty} = \lim_{n\to\infty} \frac{6}{10}$$

(252) $$\lim_{n\to\infty} \frac{\sqrt[3]{n^3 + 2n - 1}}{n+2} =$$

$$\lim_{n\to\infty} \frac{\sqrt[3]{n^3\left(1 + \frac{2}{n^2} - \frac{1}{n^3}\right)}}{n+2} =$$

$$\lim_{n\to\infty} \frac{\sqrt[3]{n^3}\sqrt{1 + \frac{2}{n^2} - \frac{1}{n^3}}}{n+2} = \lim_{n\to\infty} \frac{n \cdot \sqrt{1+0-0}}{n+2}$$

$$\lim_{n\to\infty} \frac{n \cdot 1}{n+2} = \lim_{n\to\infty} \frac{n}{n+2} = \lim_{n\to\infty} \frac{0}{0+2}$$

$$\lim_{n\to\infty} 0$$

(253) $$\lim_{n\to\infty} \frac{\sqrt[3]{n^2 + n}}{n+1} = \lim_{n\to\infty} \frac{\sqrt[3]{n^3}\sqrt{\frac{1}{n} + \frac{1}{n^2}}}{n+1} =$$

$$\lim_{n\to\infty} \frac{n\sqrt{\frac{1}{n} + \frac{1}{n^2}}}{n+1} = \lim_{n\to\infty} \frac{n^3\sqrt{0+0}}{n+1} =$$

$$\lim_{n\to\infty} \frac{0}{\infty + 1} = \lim_{n\to\infty} 0$$

254 $$\lim_{n\to\infty} \frac{\sqrt{(n^2+1+n)^2}}{\sqrt[3]{n^6+1}} = \lim_{n\to\infty} \frac{n^2+1+n}{n^2\sqrt[3]{1+\frac{1}{n^6}}}$$

$$\lim_{n\to\infty} \frac{\frac{n^2}{n^2}+\frac{1}{n^2}+\frac{n}{n^2}}{\frac{n^2\sqrt[3]{n^6+1}}{n^2}} = \lim_{n\to\infty} \frac{1+0+0}{\sqrt[3]{1+\frac{1}{n^6}}} =$$

$$\lim_{n\to\infty} \frac{1}{1} = \lim_{n\to\infty} 1$$

255 $$\lim_{n\to\infty} \frac{\sqrt{n^3-2n^2+1}+\sqrt[3]{n^4+1}}{\sqrt[4]{n^6+6n^5+2}-\sqrt[5]{n^7+3n^3+1}}$$

$$\lim_{n\to\infty} \frac{n^{\frac{3}{2}}\sqrt{1-\frac{2}{n}+\frac{1}{n^3}}+n^{\frac{4}{3}}\sqrt[3]{1+\frac{1}{n^4}}}{n^{\frac{3}{2}}\sqrt[4]{1+\frac{6}{n}+\frac{2}{n^6}}-n^{\frac{7}{5}}\sqrt[5]{1+\frac{3}{n^4}+\frac{1}{n^7}}}$$

$$\lim_{n\to\infty} \frac{\frac{n^{\frac{3}{2}}\sqrt{1-\frac{2}{n}+\frac{1}{n^5}}}{n^{\frac{3}{2}}}+\frac{n^{\frac{4}{3}}\sqrt[3]{1+\frac{1}{n^4}}}{n^{\frac{3}{2}}}}{\frac{n^{\frac{3}{2}}\sqrt[4]{1+\frac{6}{n}+\frac{2}{n^6}}}{n^{\frac{3}{2}}}-\frac{n^{\frac{7}{5}}\sqrt[5]{1+\frac{3}{n^4}+\frac{1}{n^7}}}{n^{\frac{3}{2}}}}$$

$$\lim_{n\to\infty} \frac{\sqrt{1-\frac{2}{n}+\frac{1}{n^5}} + \frac{1}{n^6}\sqrt[3]{1+\frac{1}{n^4}}}{\sqrt[4]{1+\frac{6}{n}+\frac{2}{n^6}} + \frac{1}{n^{\frac{1}{10}}}\sqrt[5]{1+\frac{3}{n^4}+\frac{1}{n^2}}} =$$

$$\lim_{n\to\infty} \frac{\sqrt{1-0+0} + 0\sqrt[3]{1+0}}{\sqrt[4]{1+0+0} + 0\sqrt[5]{1+0+0}} =$$

$$\lim_{n\to\infty} \frac{1}{1} = \lim_{n\to\infty} 1$$

256 $$\lim_{n\to\infty} \frac{\sqrt[4]{n^5+2} - \sqrt[3]{n^2+1}}{\sqrt[5]{n^4+2} - \sqrt{n^3+1}} =$$

$$\lim_{n\to\infty} \frac{n^{\frac{5}{4}}\sqrt{1+\frac{2}{n^5}} - n^{\frac{2}{3}}\sqrt{1+\frac{1}{n^2}}}{n^{\frac{4}{5}}\sqrt{1+\frac{2}{n^4}} - n^{\frac{3}{2}}\sqrt{1+\frac{1}{n^3}}} =$$

$$\lim_{n\to\infty} \frac{\frac{n^{\frac{5}{4}}\sqrt{1+\frac{2}{n^5}}}{n^{\frac{5}{2}}} - \frac{n^{\frac{2}{3}}\sqrt{1+\frac{1}{n^2}}}{n^{\frac{3}{2}}}}{\frac{n^{\frac{4}{5}}\sqrt{1+\frac{2}{n^4}}}{n^{\frac{3}{2}}} - \frac{n^{\frac{3}{2}}\sqrt{1+\frac{1}{n^3}}}{n^{\frac{3}{2}}}}$$

$\frac{3}{2}^{\cdot 10} = \frac{30}{20}$

$\frac{5}{4} = \frac{25}{20}$ $\quad \frac{-5}{20} = \frac{-1}{10}$

$$\lim_{n\to\infty} \frac{n^{-\frac{5}{10}}\sqrt{1+\frac{2}{n^5}} - n^{-\frac{5}{6}}\sqrt{1+\frac{1}{n^2}}}{n^{-\frac{7}{10}}\sqrt{1+\frac{2}{n^4}} - \sqrt{1+\frac{1}{n^3}}} =$$

$$\lim_{n\to\infty} \frac{-\infty\sqrt{1+0} - (-\infty)\sqrt{1+0}}{-\infty\sqrt{1+0} - 1} =$$

$$\lim_{n\to\infty} \frac{-\infty\cdot 1 + \infty\cdot 1}{-\infty\cdot 1 - 1} = 0$$

257 $$\lim_{n\to\infty} \frac{n!}{(n+1)! - n!} =$$

$$\lim_{n\to\infty} \frac{n!}{((n+1)\cdot n!) - n!} = \lim_{n\to\infty} \frac{n!}{n!\left((n+1)-1\right)}$$

$$\lim_{n\to\infty} \frac{n!}{n!\cdot n} = \lim_{n\to\infty} \frac{1}{n} = \lim_{n\to\infty} 0$$